HCML Monitoring Portal: Entity-Relationship Diagram (logical)

Alerting and availability

TRIGGER · triggers			
string	triggerid	PK	
string	manual_close		1 Close permitted

objectid when object=0

PROBLEM · problem			
string	eventid	PK	
string	objectid	FK	the trigger, no constraint in MySQL
string	name		the notification text
string	severity		0 to 5
number	clock		
string	r_eventid		0 means still open
string	acknowledged		

derives, clipped to the window

one call per problem, cached

INCIDENT			
string	eventid	PK	
number	start		clipped to the window
number	end		recovery clock, or now
boolean	resolved		

AI_EXPLANATION	
string	summary
string	businessImpact
string	recommendation

merged intervals

HOST_AVAILABILITY			
string	hostid	PK	
number	availability		percent, overlaps merged
number	downtime		
number	incidents		
boolean	measured		false when nothing was collected

selectProblemEvents, the root causes

rolled up per site

selectHosts, the portal keeps the first

Services and SLA

SLA · sla			
string	slaid	PK	
string	name		
number	slo		target percent
string	period		0 daily to 4 annually

sla.getsli, positional

SLI · computed, no table		
number	sli	achieved percent
number	uptime	
number	downtime	
number	error_budget	seconds, negative means missed

serviceids index

SERVICE · services			
string	serviceid	PK	
string	name		
number	status		-1 OK, else severity
string	description		

selectChildren, a DAG not a tree

1 to 3 icmpping items collapse into one link

Hosts, sites and links

LINK			
string	id	PK	hostid:target
string	target		
number	loss		
number	latency		
string	state		up, degraded, down, unknown

SITE			
string	name	PK	derived, never stored
string	code		from the host-name convention
number	worst		highest severity, -1 when clear
number	total		

HOST · hosts			
string	hostid	PK	
string	name		the visible name
string	status		0 monitored, 1 not monitored
string	maintenance_status		1 in maintenance

resolveSite(), a total function

derives

SITE_HOST			
string	hostid	PK	
string	state		up, down, degraded, nodata, disabled
string	reason		why, ping first
string	siteSource		tag, name, inventory or group

selectInventory, [] when disabled

HOST_INVENTORY · host_inventory		
string	site_city	site signal, rank 3
string	location	site signal, fallback
string	poc_1_name	owner signal
string	poc_1_email	owner signal

selectInterfaces

HOST_INTERFACE · interface		
string	ip	
string	type	1 agent, 2 SNMP, 3 IPMI, 4 JMX
string	available	0 unknown, 1 up, 2 down
string	main	1 the default interface

selectTags

HOST_TAG · host_tag		
string	tag	site, owner, criticality
string	value	

selectHostGroups

HOST_GROUP · hstgrp		
string	groupid	PK
string	name	

hostids

Metrics

ITEM · items			
string	itemid	PK	
string	key_		parsed for icmpping targets
string	name		
string	value_type		picks the history and trend table
string	lastvalue		not a column, taken from history
string	lastclock		blank beyond the history period

itemids

TREND · trends, trends_uint			
string	itemid	FK	no constraint in MySQL
number	num		samples in that hour
number	value_avg		
number	value_max		

itemids

HISTORY · five history tables			
string	itemid	FK	no constraint in MySQL
number	clock		
number	value		